Table 7. Data and compared methods.

Data	
Registry	569 tools, 76 servers (LiveMCPTool [1] + MCPTool-Bench++ [2]); 44 annotated
Multi-step tasks	95 LiveMCPBench tasks (test); 60 development prompts
Single-tool requests	60 development / 300 test (50 per category, 66 targets)
Calibration suite	432 items: 6 operations, $n \in \{10, 40, 160, 640\}$, en/fr, noise, context; 264 cal. / 168 test
Executed servers	6 MCP reference servers, 37 registry tools
Synthetic	cloned and heterogeneous registries (10^3–10^5); 80 cover and 120 QoS instances
Baselines (adapted to one registry, one encoder; tuned on the development split)	
RAG-MCP [3]	dense top-k over the request
MCP-Zero [4]	model-written requests, server routing
COLT [5]	collaborative graph of 450 training queries
ToolScope [6]	LLM-confirmed tool merging, subquery retrieval
CMTF [7]	typed-state search, one tool per step
BM25, RRF, per-subtask	lexical, fused, one tool per analyst subgoal
Ablations of PaMCP	
G, P, S	gap (Alg. 1), eligibility (Sec. 5), set reasoning + certificate (Sec. 6), 2^3 design
PaMCP-L	relevance-first ranking selected on the development split

Table 8. Parameters varied (default in bold).

Parameter	Values
Profile threshold τ / risk target	0.6, 0.75, **0.9** / 0.05, 0.10
Forging servers δ; false accepts φ	0–0.3 (5 seeds); 0–0.2, **0.05**
Probe coverage κ; trust $\ell_{\min}$	0.25–**1**; 0.2–0.95, **0.65**
Candidate budget K; size matching	1, 5, **10**, 20, 40; equal $\lvert K\rvert$
Poisoned descriptions; policies	10%, 30%; least privilege, read-only
Registry size; pair relation	569–10^5; eager, partitioned, **lazy**
Cost weights; utility floor $u_{\min}$	0, 0.25, **0.5**, 1; 0–0.9

8 Experimental Evaluation

PaMCP claims that discovery can be conditioned on the deployed model (RQ1), restricted to tools whose safety-relevant behaviour is evidenced (RQ2), and returned as a small toolkit whose interfaces, effects and flows are certified jointly before any call (RQ3), at a cost that scales and under registry change (RQ4). This section tests each claim against the strongest baselines we could configure, labels every result by its evidence class (*measured* from the running system, *observed* from executed servers, *labelled* against reference labels, *simulated* under stated assumptions), and confirms the main comparisons with pre-registered tests.

8.1 Experimental Setup

Table 7 lists the data and the methods and Table 8 the varied parameters. Effects and types are scored against blind labels J_B (Claude Opus) that are not used to simulate evidence; the simulated verifier uses labels J_A (Claude Sonnet), and we report the swapped direction and executed servers to remove the dependency between the two. All LLM outputs were recorded once and are replayed. Every method, including PaMCP, was configured on the development split only. Request-level metrics are resampled with target tools as clusters and task-level metrics with tasks; paired tests are sign-flip permutation tests and proportions use Wilson intervals (95%). Two confirmatory families (F1–F7, H1–H5) were fixed before the data they test and are Holm-corrected. Code, records, manifests and a notebook that re-runs every experiment on a CPU are public.[1]

Table 9. Model-profile calibration: accuracy (Wilson 95% CI), self-assessment calibration, and the held-out behaviour of the threshold τ chosen for a 5% risk target (measured profile) versus the 64-item profile at $\tau = 0.9$. Cov.: share of operations kept native; FNR: false-native rate (errors among them).

				Self-assess.		Measured, risk 5%			64-item	
Model	n	Accuracy (%)	n=640	ECE	Brier	τ	Cov.	FNR	Cov.	FNR
Haiku	432	86.1 [82.5, 89.1]	67	0.39	0.30	0.21	83	3.6	21	0.0
Sonnet	432	91.4 [88.4, 93.7]	62	0.20	0.11	0.42	90	3.3	26	4.5
Opus	387	100.0 [99.0, 100.0]	100	0.38	0.20	0.90	100	0.0	–	–

8.2 RQ1: Model-Conditioned Representation

Question. Does a measured profile ρ_{LLM} decide reliably which operations the deployed model keeps, and does the gap respond to the model and the input? *Baselines:* the model's own self-assessment and the earlier 64-item profile. *Protocol:* three models answer the calibration suite by reading only, after stating per cell their probability of success; τ is chosen on the calibration split for a risk target and evaluated on held-out items (measured).

Results. Capability boundaries differ by model and operation (Table 9): Haiku fails counting from $n = 40$, Sonnet loses joins as they grow, and both fail approximate sums over 640 values; Opus answers all 387 items it was given. With the measured profile and a 5% risk target, the held-out false-native rate stays within the target for every model (3.6%, 3.3%, 0.0%) while 83–100% of the operations remain with the model (Fig. 8b). Both baselines fail: self-assessment is miscalibrated for every model (ECE 0.20–0.39, all underconfident; Fig. 8a), and the 64-item profile keeps only 21–26% of the operations native, sending most correctly performed work to tools. The gap behaves as designed (Fig. 8c): two tools suffice for 30 or 300 reviews, six for 3,000, and the engine abstains rather than return an unsupported toolkit at 30,000; without gap analysis it abstains in every setting. On the test tasks the profile overrules 9.4% of the analyst's claims that an operation is native. *Answer:* a measured, risk-targeted profile makes model conditioning reliable; neither self-assessment nor a small profile can replace it.

8.3 RQ2: Trust under Missing, Forged and Imperfect Evidence

Question. Does provenance-sound eligibility (P) keep unsafe tools away from the model, and at what cost? *Baselines:* no filter, MCP's default (annotations displayed, not enforced), annotations as a closed world, and lexical cues, each combined with PaMCP's retrieval and with MCP-Zero. *Protocol:* read-only policy over 273 requests with read-only gold tools; Exp@5 is the share whose top five contain a tool that is not read-only under J_B, Util@5 the share containing a gold tool; forgery and verifier errors are simulated.

[1] `https://github.com/msellamiTN/taas-provenance-discovery`, branch `research/pamcp-final-evaluation`; guide in `docs/REPRODUCIBILITY.md`.

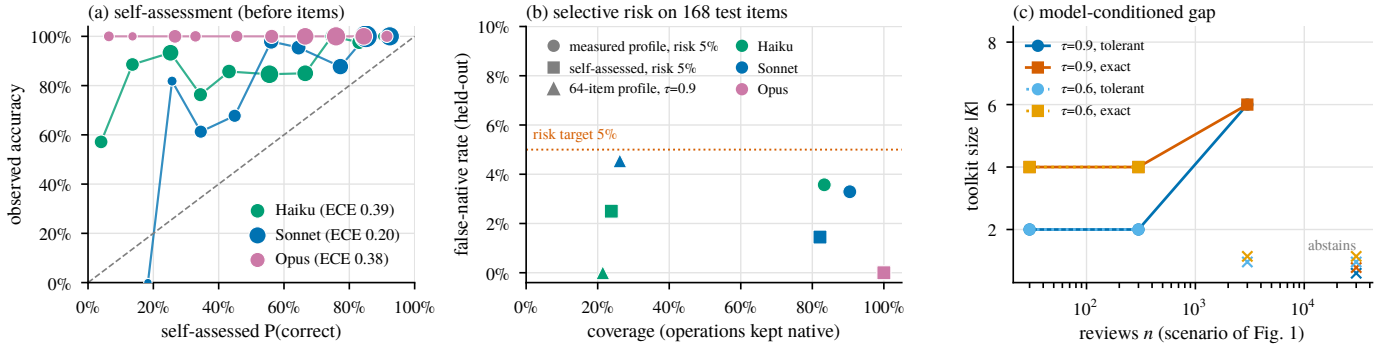


Figure 8. RQ1. (a) Reliability of the models' self-assessment. (b) Held-out coverage and false-native rate of the measured profile against the two baselines. (c) Toolkit size of the review scenario against the input size (crosses: abstentions).

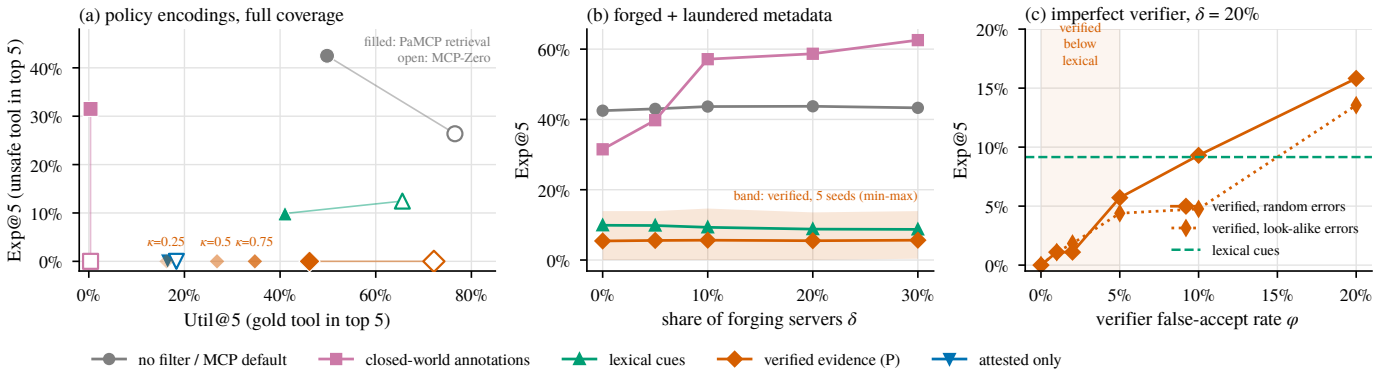


Figure 9. RQ2. (a) Exposure against utility per policy encoding (filled: PaMCP retrieval, open: MCP-Zero) and under lower probe coverage κ. (b) Exposure as a share δ of servers forges annotations and launders descriptions (band: five seeds). (c) Exposure against the verifier's false-accept rate.

Table 10. Encodings of a read-only policy over 273 requests with read-only gold tools: eligible share of the registry, unsafe tools admitted (of 120, J_B), exposure and utility of the top five for PaMCP's retrieval stage and MCP-Zero, and exposure when 20% of the servers forge and launder their metadata (verifier false-accept rate 5%, mean of five seeds).

			PaMCP retr.		MCP-Zero		δ=0.2
Encoding	Elig.	Unsafe	Exp@5	Util@5	Exp@5	Util@5	Exp@5
no filter	100.0	120	42.5	49.8	26.4	76.6	43.7
MCP default (display only)	100.0	120	42.5	49.8	26.4	76.6	43.7
closed-world annotations	7.7	1	31.5	0.4	0.0	0.4	58.7
lexical cues	67.8	21	9.9	41.0	12.5	65.6	8.8
verified evidence (P)	77.0	0	0.0	46.2	0.0	72.2	5.5
observed + lexical fallback	77.0	0	0.0	46.2	0.0	72.2	5.5
attested only	26.4	0	0.0	16.5	0.0	18.3	5.8

Results. Every baseline encoding exposes unsafe tools (Table 10): 42.5% of the requests without a filter or under MCP's default, 31.5% with annotations as a closed world (whose utility falls to 0.4%), and 9.9% with lexical cues. Verified evidence admits none of the 120 unsafe tools, keeps 77% of the registry eligible and costs 3.6 utility points (4.4 for MCP-Zero); the difference to lexical cues is confirmatory (F4: -9.9 points, $p_{\text{Holm}} < 0.001$). Under forgery the closed-world reading collapses (31.5% to 62.6% exposure at $\delta = 0.3$) while verified evidence stays at 5.4–5.6% for every δ with a 5% false-accept rate (Fig. 9b), and it remains below lexical cues up to $\varphi = 5\%$ (10% for look-alike errors; Fig. 9c). Three checks remove the dependency on how the labels were built. With the verifier simulated from J_B and scored on J_A, exposure is 2.3–3.9% across PaMCP, RAG-MCP, MCP-Zero and ToolScope, a quarter to two fifths of the exposure left by lexical cues. Executing the six MCP reference servers under system-call tracing (observed; Fig. 10), both labellings agree with all 30 completed calls: none of the 13 tools observed to change state is labelled read-only (Wilson upper bound 22.8%). And every one of the 37 surviving definitions changed since the benchmark snapshot, so evidence not bound to $h(t)$, as PaMCP binds it, would have been stale. *Answer:* eligibility based on verified, hash-bound evidence removes the exposure that published metadata leaves, degrades in proportion to verifier error, and does not depend on the reference labels.

8.4 RQ3: Composition and Certification

Question. Does set-level reasoning return smaller toolkits that are jointly feasible more often than retrieval? *Baselines:* RAG-MCP, COLT, ToolScope (top five), MCP-Zero, CMTF and per-subtask retrieval, each also restricted to eligible tools. *Protocol:* 94 test tasks; group recall against benchmark gold (measured); joint feasibility decided by two evaluators written independently of the certificate, a coarse one and a strict one that also checks domains, cardinality, schema-required inputs and egress of private data (labelled). On ten hand-built cases with known answers the strict evaluator decides 7 correctly, the coarse one 2.

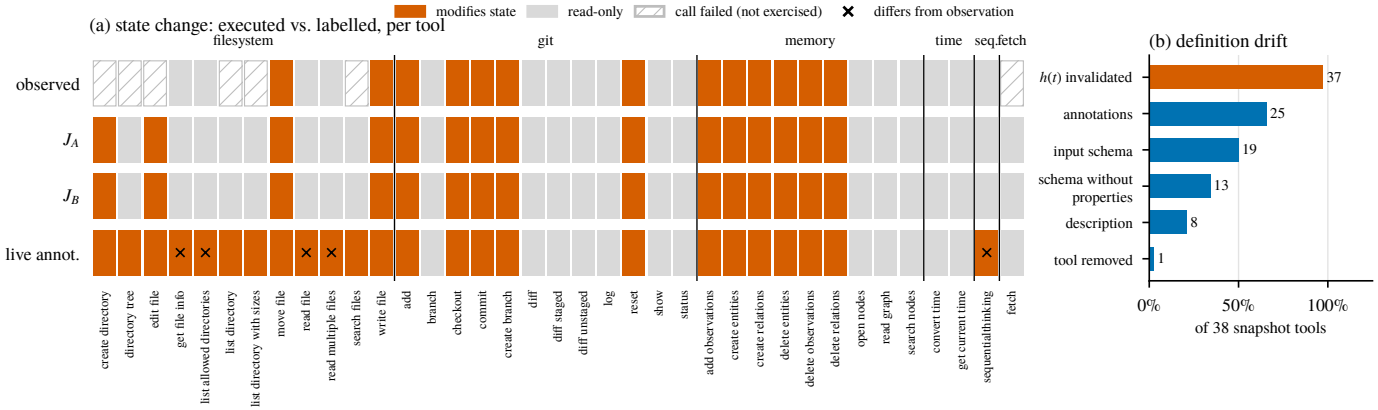


Figure 10. RQ2, executed reference servers (observed). (a) Per registry tool: state change observed under tracing against the two labellings and the servers' current annotations. (b) Changes of the live definitions since the benchmark snapshot; every change alters $h(t)$ and invalidates bound evidence.

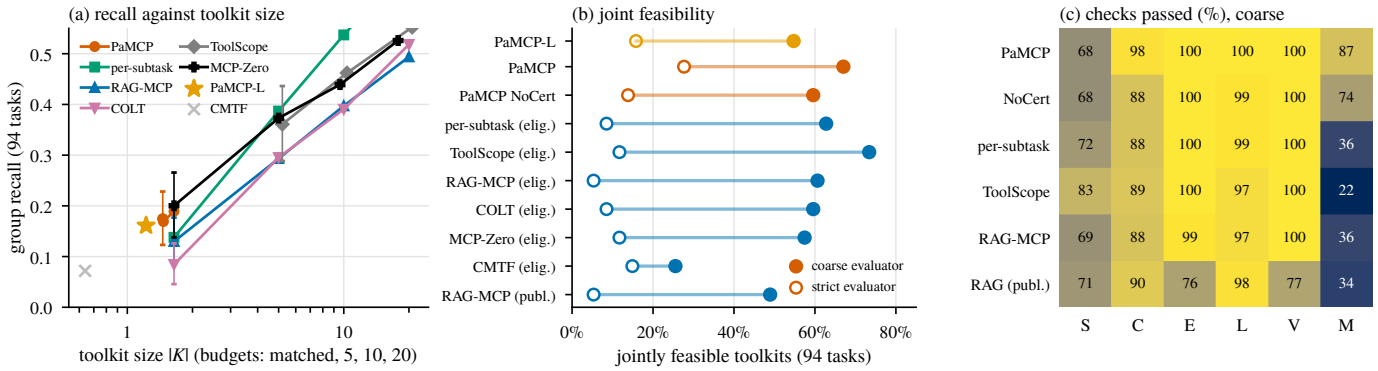


Figure 11. RQ3. (a) Group recall against toolkit size for candidate budgets. (b) Jointly feasible toolkits under the coarse and the strict evaluator. (c) Checks passed: goals served (S), inputs obtainable (C), effects (E), flows (L), evidence (V), minimality (M).

Table 11. Toolkits on the 94 LiveMCPBench test tasks: size, manifest tokens, group recall (task bootstrap 95% CI), share of toolkits jointly feasible for the coarse and the strict independent evaluator (least-privilege policies), and abstentions. Baselines restricted to eligible tools unless marked published.

				Joint feas. (%)		
Method	$\|K\|$	Tok.	Group recall [CI]	coarse	strict	Abst.
PaMCP (K=10)	1.46	198	0.174 [0.12, 0.23]	67.0	27.7	7.4
PaMCP-L	1.22	146	0.161	54.7	15.8	9.5
PaMCP NoCert	–	–	–	59.6	13.8	0.0
Per-subtask (elig.)	2.59	346	0.249 [0.19, 0.32]	62.8	8.5	–
RAG-MCP@5 (elig.)	5.00	1244	0.293 [0.22, 0.37]	60.6	5.3	–
COLT@5 (elig.)	5.00	1094	0.295 [0.23, 0.37]	59.6	8.5	–
ToolScope@5 (elig.)	5.20	976	0.361 [0.29, 0.44]	73.4	11.7	–
MCP-Zero (elig.)	1.76	250	0.227 [0.17, 0.29]	57.4	11.7	–
CMTF (elig.)	0.64	105	0.072 [0.04, 0.11]	25.5	14.9	–
RAG-MCP@5 (published)	5.00	1244	0.293 [0.22, 0.37]	48.9	5.3	–

Table 12. Single-tool retrieval on the 300 test requests with configurations selected on the development split: Hit@1 over the 81 MCPToolBench++ tools and, with target-tool clustered 95% CI, over the 569-tool registry, and the other metrics over 569 tools.

	81 tools	569 tools				
Method	Hit@1	Hit@1 [CI]	Hit@3	Hit@5	MRR	nDCG
BM25	52.7	41.0 [26, 60]	57.3	63.7	0.503	0.543
RAG-MCP (dense)	49.7	37.3 [22, 56]	50.0	54.3	0.451	0.492
RRF fusion	54.3	38.7 [23, 57]	54.7	61.3	0.481	0.525
MCP-Zero	68.0	66.7 [49, 80]	80.0	85.7	0.743	0.775
COLT	49.7	37.3 [22, 56]	50.0	54.3	0.451	0.492
ToolScope	53.0	41.0 [26, 60]	58.3	63.7	0.504	0.543
CMTF	58.7	32.7 [19, 49]	47.0	54.7	0.428	0.488
PaMCP (Eq. 5)	61.0	29.7 [16, 47]	45.7	53.7	0.400	0.461
PaMCP-L	72.0	41.3 [25, 60]	61.3	69.3	0.536	0.603
PaMCP, oracle types (diag.)	71.7	58.3 [38, 74]	75.3	81.3	0.680	0.728

Results. PaMCP returns the smallest toolkits apart from CMTF, 1.46 tools and 198 tokens against 5 tools and 976–1,244 tokens for the top-five retrievers (Table 11), and it certifies 92.6% of the tasks, abstaining with a counterexample on the rest. Under the strict evaluator its toolkits are jointly feasible in 27.7% of the tasks, 1.9 to 5.2 times as often as every baseline restricted to eligible tools (5.3–14.9%) and twice as often as without certification (13.8%; Fig. 11b). Certification acts through obtainable inputs and minimality (C 88% to 98%, M 74% to 87%, against 22–36% for the top-five retrievers; Fig. 11c). At equal size PaMCP reaches more gold steps than RAG-MCP, per-subtask retrieval and COLT (0.189 against 0.130, 0.138 and 0.084; F2 not significant after correction), whereas retrievers that expose three to four times more tools reach more (0.249–0.361; non-inferiority to per-subtask retrieval, F1, not shown): PaMCP trades breadth for certified compactness.

Boundary: single-tool retrieval. Retrieval is not where PaMCP gains (Table 12). Over 81 tools its relevance-first ranking is the most accurate (72.0% Hit@1, MCP-Zero 68.0%), but over the full registry MCP-Zero leads (66.7%), PaMCP-L ties

Table 13. Mechanism ablation on 155 tasks under least-privilege policies: the 2^3 design over the gap (G), provenance-sound eligibility (P) and set-level reasoning (S), and five components of the full configuration. Reach: task–attack pairs whose out-of-scope action the toolkit can realise.

G	P	S	Recall	Tokens	Joint (%)	Reach (%)	Abst. (%)
–	–	–	0.270	297	53.5	1.1	0.0
on	–	–	0.239	208	55.5	0.4	0.0
–	on	–	0.295	316	58.1	0.1	0.0
–	–	on	0.223	221	52.9	0.1	16.1
on	on	–	0.253	229	56.8	0.1	0.0
on	–	on	0.230	189	54.8	0.0	6.5
–	on	on	0.183	193	55.5	0.0	20.0
on	on	on	0.181	176	56.1	0.0	10.3
Full, no backward closure			0.181	173	53.5	0.0	12.3
Full, flow along derivations			0.236	206	60.6	0.0	5.8
Full, no relevance term			0.167	141	54.8	0.0	10.3
Full, no utility floor			0.181	176	56.1	0.0	10.3
Full, no refinement			0.179	175	55.5	0.0	11.0

tuned BM25 (41.3% against 41.0%; H1 n.s.) and the structural ranking of Eq. (5) falls to 29.7% (F7 against PaMCP), because types agree with J_B on 78.9% of output kinds but only 35.7% of domains. With correct types the same ranking reaches 58.3%, above every baseline except MCP-Zero, which locates the limit in the typing rather than in the composition. *Answer:* set-level reasoning yields the most compact and most often jointly feasible toolkits; finding more tools requires better typing or a larger budget.

8.5 RQ2/RQ3: Security and Contribution of the Mechanisms

Question. Does the exposed toolkit limit what an injected instruction could achieve, and which mechanism is responsible? *Protocol:* for 155 tasks and each attack class outside the task's needs, an attack is reachable if the toolkit can realise it under J_B (labelled, offline); poisoning places a canary in the description, name or argument keys of 10–30% of the tools and scans what each method shows the model; the mechanisms are varied in a 2^3 design with Shapley attribution.

Results. No out-of-scope action is reachable through PaMCP's toolkits: 0 of 712 task–attack pairs under least privilege and 0 of 775 under a read-only policy, against 0.7–8.1% and 8.4–14.7% for the published baselines (Fig. 12a; F5, $p_{\text{Holm}} < 0.001$). Counting only attacks that are definitely out of scope, no PaMCP toolkit admits one, against 15.8% of the tasks for MCP-Zero, 20.0% for per-subtask retrieval, 33.7% for RAG-MCP and 37.9% for ToolScope (H5: -20.0 points, 95% CI $[-28.4, -12.6]$, $p_{\text{Holm}} < 0.001$); 32.6% admit an attack whose scope the labels cannot decide. With 30% of the descriptions poisoned, the aliased manifest never shows an injected instruction, whereas the definitions exposed by the baselines carry one in 17.4–86.2% of the tasks (Fig. 12b); 55.5% of PaMCP's toolkits return remote content, but none of them also enables an out-of-scope action (3.2–22.6% for the baselines).

The factorial design (Table 13) shows that each mechanism contributes. Eligibility removes most reachable actions, and with gap analysis and set reasoning reachability falls from 5.2% of the tasks to none (Shapley shares 23%, 35% and 42% for G, P and S). Gap analysis and set reasoning shrink the context from 297 to 176 tokens (main effects -56 and -67 tokens, $p < 0.001$), and the gap halves abstentions (10.3% against 20.0%). The price is recall (0.270 to 0.181, mainly through S); no mechanism changes the coarse joint feasibility. *Answer:* under the labelled threat model, PaMCP's mechanisms jointly close the action and metadata channels that retrieval baselines leave open.

8.6 RQ3/RQ4: Efficiency, Optimality and Evolution

Question. What does discovery cost, how close is the composition to the optimum, and does the engine keep its guarantees under change? *Baselines:* eager materialisation of the risk relation, the greedy regime and an integer program (HiGHS) written independently of the engine, full rebuilds and cache flushes, and cold discovery without experience reuse.

Results. On the real registry a request takes 49 ms at the median (eligibility 0.08 ms; RAG-MCP's single dense ranking 12 ms) and 2.7 s at 10^5 cloned tools. Deciding the pairwise risk relation lazily or by effect class gives identical toolkits on all test tasks and builds the engine at 10^5 tools in 133–134 s instead of 2,224 s, 16.5 times faster with near-linear growth (Fig. 13a); on heterogeneous registries of 10^4 tools the lazy build takes 4.6–5.9 s. Wherever the exact search finishes it returns the ILP optimum (61 of 61 solved instances across alternatives, depth, arity, conflicts and shared tools), it recognises all ten infeasible instances, and the greedy regime is 4.4–12.6% above the optimum on average (Fig. 13b); with six goals the search exhausts its budget and falls back to greedy (6.9% above). When providers differ in price, latency and reliability, every equal-weight choice is Pareto-optimal and meets the utility floor, and weights move the choice along the front as intended (price only: $0.33\times$ price; latency only: $0.73\times$ latency; Fig. 13c). Under change, an incremental update takes 13.5 ms against 4.2 s for a rebuild, the exact invalidation of Proposition 3 touches 1.1% of the cached toolkits instead of all, and all 20 simulated rug pulls lose their evidence. Experience reuse serves 32.7% of 300 requests without an analyst call, with unchanged certification (82.7% against 82.3%) and no violation in 89 revalidation probes (removed servers, revoked or rewritten tools, stricter principal, other model). Deterministic repair raises certification from 72.6% to 87.4% when counterexamples are induced. *Answer:* the complete engine scales to 10^5 tools, composes optimally where it searches exactly, makes Pareto-consistent QoS trade-offs and adapts to change without losing its guarantees.

8.7 Summary and Threats to Validity

Table 14 summarises the answers and Table 15 the confirmatory tests. All confirmatory tests on safety (F4, F5, H5) favour PaMCP at $p_{\text{Holm}} < 0.001$; the tests on recall and retrieval do not, and the feasibility advantage under the strict evaluator is exploratory. *Validity.* Effects, types and feasibility are scored against LLM labels; execution confirms them on 30 calls of six non-adversarial reference servers, and the label swap bounds the effect of labeller disagreement but not of errors shared by one model family. Reachability is decided offline: whether an agent follows an injected instruction, and end-to-end task success, require executing the toolkits, which this study did not do.

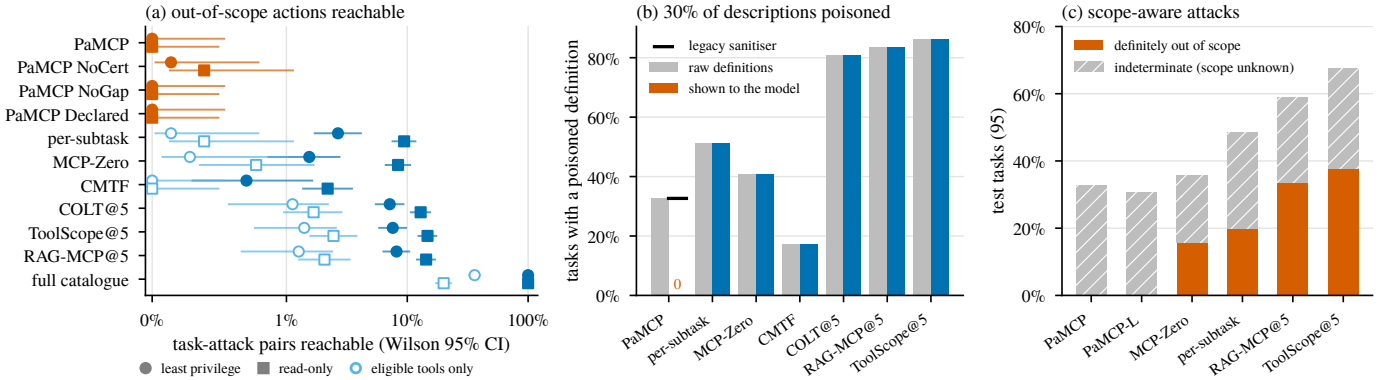


Figure 12. Security. (a) Task–attack pairs reachable for each method, as published (filled) and restricted to eligible tools (open), under two policies (Wilson 95% CI). (b) Tasks whose definitions carry an injected instruction, raw against shown to the model. (c) Test tasks with an attack definitely out of scope or of undecidable scope.

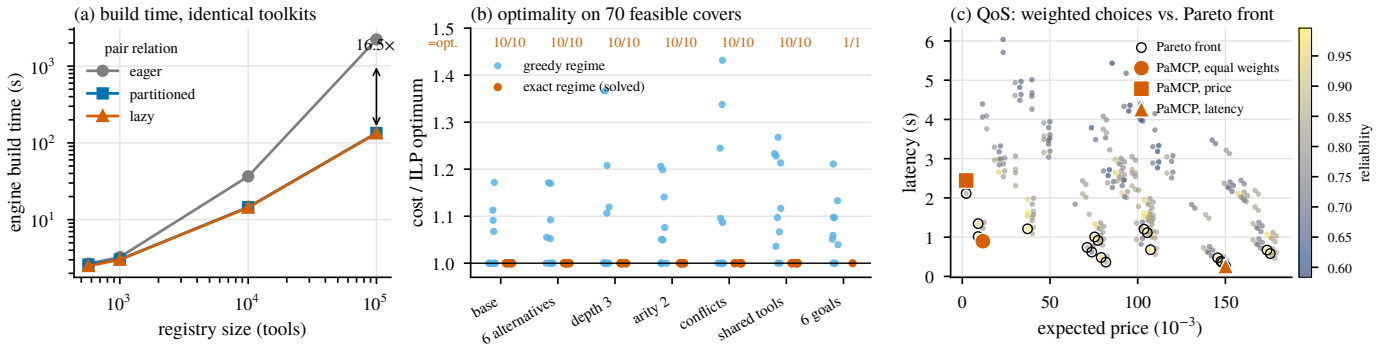


Figure 13. RQ3/RQ4. (a) Engine build time by pair-relation mode (identical toolkits on all test tasks). (b) Cost of the greedy and the exact regime relative to the ILP optimum (top: exact equals optimum / solved). (c) The 256 toolkits of a QoS instance, their Pareto front and PaMCP's choices under three weightings.

Table 14. Answers to the research questions. Conf.: confirmatory test (p_{Holm}).

RQ	Result	Best baseline	Conf.
1	FNR ≤3.6% at 83–100% coverage	self-assessment ECE 0.20–0.39	–
2	Exp@5 0% (swap 2.3–3.9%), Util −3.6	lexical 9.9%, MCP default 42.5%	F4 <0.001
2	labels = execution, 30/30 calls	– (observed)	–
3	strict joint 27.7%, 1.46 tools	5.3–14.9%, 0.64–5.2 tools	expl.
3	recall at equal size 0.189	0.084–0.200	F2 n.s.
3	Hit@1 (569) 41.3%	MCP-Zero 66.7%	F7 against
2/3	reachable 0/712, definite 0%	0.7–8.1%, 15.8–37.9%	F5, H5 <0.001
4	build 16.5× faster; =ILP 61/61	eager 2,224 s; greedy +4–13%	–
4	update 13.5 ms; reuse 32.7%	rebuild 4.2 s; cold	–

Baselines are adaptations sharing a TF-IDF and SVD encoder, all tuned on the same development split; larger registries are cloned or generated and QoS data are synthetic. Executed compositions over isolated and adversarial servers, human annotation of a stratified tool sample and reproductions of the baselines with their original encoders are the natural next steps, and the released artifact supports each of them.

Table 15. Confirmatory comparisons, fixed before the data they test (F1–F7 before the set-level experiments; H1–H5 in a dated amendment before the revised test runs), Holm-corrected within each family. Differences in points (tokens for F6), PaMCP minus comparator, with 95% CI.

	Comparison	Δ	95% CI	p_{Holm}	Verdict
F1	Group recall vs per-subtask (non-inf., 0.05)	-8.0	[-16.6, +0.2]	1.000	not shown
F2	Size-matched recall vs RAG-MCP	+6.0	[-1.6, +13.7]	0.397	n.s.
F3	Joint feasibility vs per-subtask (elig.)	+4.3	[-8.5, +18.1]	1.000	n.s.
F4	Exp@5 verified vs lexical	-9.9	[-13.9, -6.6]	<0.001	supports
F5	Reachable attacks vs per-subtask	-2.7	[-3.9, -1.5]	<0.001	supports
F6	Tokens, full vs NoGap	-16.7	[-34.6, +1.0]	0.269	n.s.
F7	Hit@1 goal contract vs raw RRF (569)	-15.7	[-21.3, -10.0]	<0.001	against
H1	Hit@1 PaMCP-L vs tuned BM25 (569)	+0.3	[-8.9, +8.8]	1.000	n.s.
H2	Hit@1 PaMCP-L vs Eq. (5) (569)	+11.7	[+3.0, +22.6]	0.070	n.s.
H3	Group recall PaMCP-L vs per-subtask (non-inf.)	-8.8	[-17.6, -0.5]	1.000	not shown
H4	Strict joint feas. vs per-subtask (elig.)	+7.4	[+0.0, +14.9]	0.277	n.s.
H5	Definite out-of-scope vs per-subtask	-20.0	[-28.4, -12.6]	<0.001	supports

REFERENCES

[1] G. Mo, W. Zhong, J. Chen, Q. Yuan, X. Chen, Y. Lu, H. Lin, B. He, X. Han, and L. Sun. LiveMCPBench: Can agents navigate an ocean of MCP tools? *arXiv preprint arXiv:2508.01780*, 2025.

[2] S. Fan, X. Ding, L. Zhang, and L. Mo. MCPToolBench++: A large scale AI agent model context protocol MCP tool use benchmark. *arXiv preprint arXiv:2508.07575*, 2025.

[3] T. Gan and Q. Sun. RAG-MCP: Mitigating prompt bloat in LLM tool selection via retrieval-augmented generation. *arXiv preprint arXiv:2505.03275*, 2025.

[4] X. Fei, X. Zheng, and H. Feng. MCP-Zero: Active tool discovery for autonomous LLM agents. *arXiv preprint arXiv:2506.01056*, 2025.

[5] C. Qu, S. Dai, X. Wei, H. Cai, S. Wang, D. Yin, J. Xu, and J.-R. Wen. Towards completeness-oriented tool retrieval for large language models. In *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management (CIKM '24)*, pages 1930–1940. ACM, 2024.

[6] M. M. Liu, D. Garcia, F. Parllaku, V. Upadhyay, S. F. A. Shah, and D. Roth. ToolScope: Enhancing LLM agent tool use through tool merging and context-aware filtering. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 34095–34119, 2026.

[7] R. Suresh Babu and L. Ganesh Iyer. ToolChoiceConfusion: Causal minimal tool filtering for reliable LLM agents. arXiv:2606.06284, 2026.